

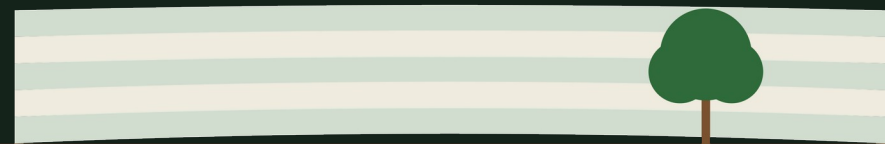
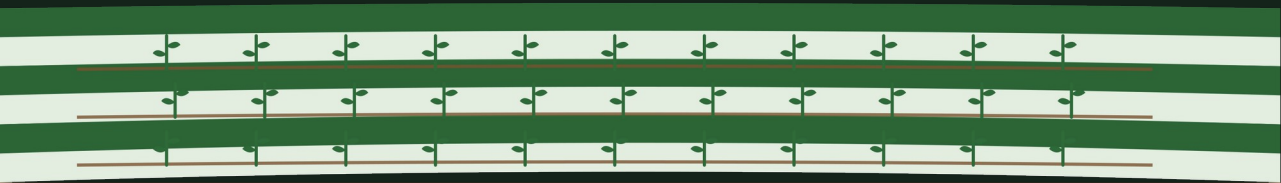
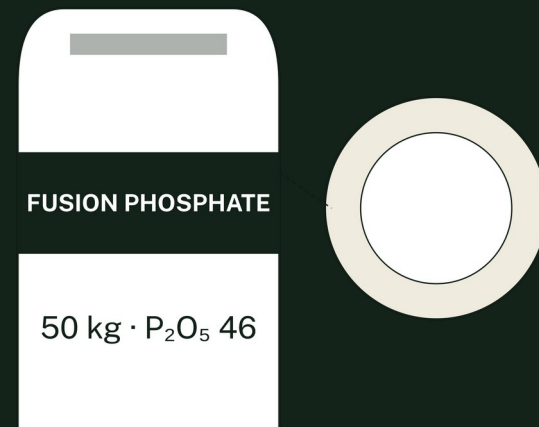


DEALER TRAINING · SEPTEMBER 2026

Fusion Phosphate

Triple super phosphate, P_2O_5 46%, no nitrogen, 50 kg, drilled into the ground at sowing.

For VAN dealers and the field team, written so a dealer can repeat it to a grower. Lahore.



What a dealer says in 30 seconds.

Fusion Phosphate is triple super phosphate: P_2O_5 46%, 43 of the 46 water soluble, no nitrogen, 50 kg, granular. Phosphorus does not travel to the root; the root has to grow to it. So this bag goes in the ground before the crop, at land preparation or sowing, drilled beside and below the seed row where the equipment allows. The granule dissolves into an acidic solution around itself, the opposite of the calcareous soil it sits in, which gives the root a few weeks to reach it before the soil's calcium takes it back. VAN sells 2 basal phosphates. This one is the straight, high-analysis bag with no nitrogen. Green Phosphate is the coated bag with the humic seal. The programme's bag count is the same for either. The grower chooses.

The 4 things it must contain

1

The bag: P_2O_5 46%, 43% water soluble, no nitrogen, 50 kg, PS 216-2009. 23 kg of P_2O_5 in every bag.

2

The nutrient: phosphorus builds the root in the first weeks and does not travel. It must be in the ground before the crop.

3

The method: drill at sowing, beside and below the seed. Side-dress second. Broadcast last.

4

The choice: Fusion Phosphate or Green Phosphate at land preparation, same bag count in the programme. No trial says one beats the other, and VAN says so.

Say it in the grower's own words. Every figure in these 30 seconds is on the bag, in the published crop programme, or in the named evidence on the slides that follow.

THE BAG

What the bag says, and what it is registered as.

50 kg

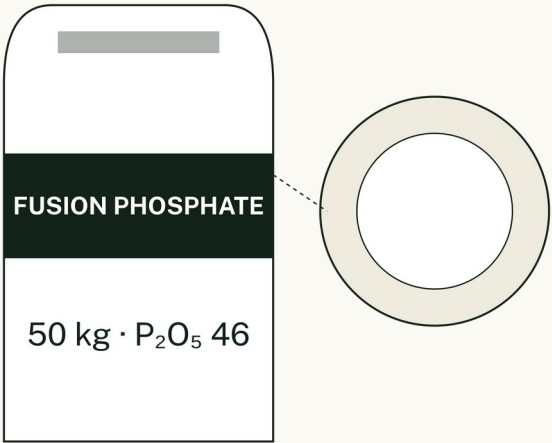
the bag. 23 kg of P₂O₅ in every bag, from the declared analysis. No nitrogen.

P₂O₅ 46%

total phosphate, declared on the front. The highest analysis a straight phosphate carries.

43%

of the bag is water-soluble P₂O₅: 21.5 kg of the 23 kg dissolves at once, as van.com.pk prints it.



pH 2.5 to 3.5

the reaction of the granule in solution, from VAN's product data. Acidic, the opposite of the soil.

1 to 4 mm

granule size, sized for the seed drill's fertilizer box and for even spreading.

LAB 336

every batch is released through VAN Lab, PNAC accredited, before it ships.

Registered as	Pakistan Standard	Manufacturing licence
Fusion Phosphate	PS 216-2009	CM/L-2942/2023

No expiry date is printed in this deck. The licence number is what a dealer quotes.

A GROWER'S QUESTION

**“I put my phosphate in once, at sowing, and never again.
Which bag, and what am I actually paying for?”**

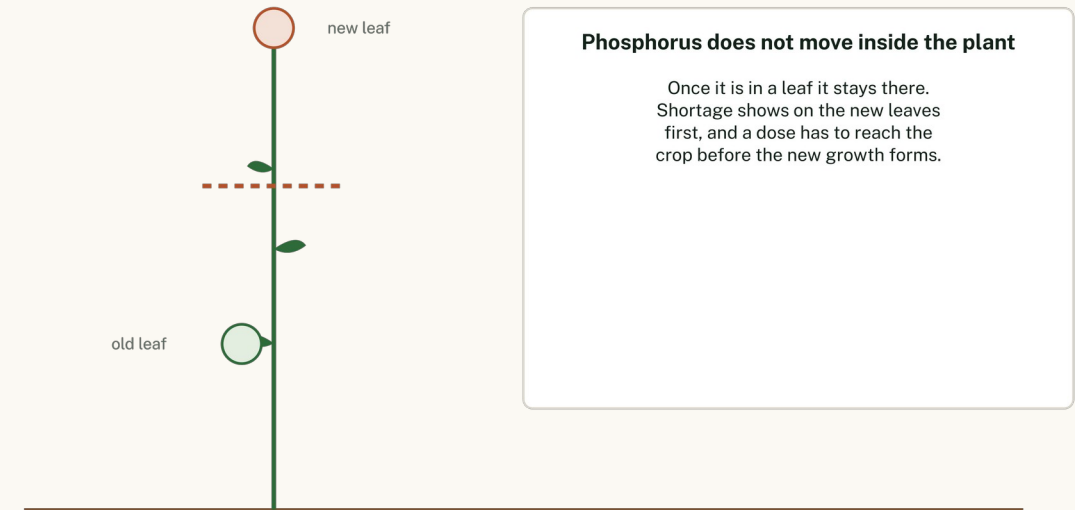
The grower has 1 pass at land preparation and 2 VAN bags to choose from. He wants to know what is in each, why it must go in before the crop, and why the kilograms on the bag are not the whole answer. The answer is 3 ideas: phosphorus does not travel, the granule makes its own acidic zone, and the 2 basal bags are a choice, not a contest.

The 3 answers on the next 3 slides, 1 idea each.

Phosphorus does not travel. The root has to reach it.

Nitrogen moves with the water. Phosphate barely moves at all: in soil it travels millimetres, so a root has to grow to the granule. That is why phosphorus is a sowing-time nutrient. It builds the root in the first weeks, decides how many tillers wheat makes and how the crop sets and fills, and most of that damage, if it is short, is done early and cannot be undone later. A bag put on a standing crop is a bag the root may never meet.

Say it like this: “Phosphorus does not come to the root. The root has to go to it. So it goes in the ground before the crop.”

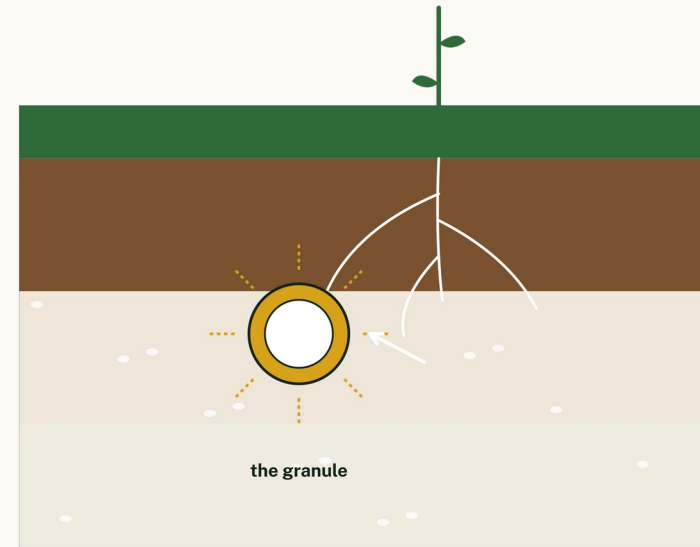


Mobile against immobile. A picture of the idea, general soil chemistry, no VAN measurement.

The granule makes its own acidic zone, for a few weeks.

Triple super phosphate is a calcium phosphate made with phosphoric acid. When water reaches the granule it dissolves into a strongly acidic solution, pH 2.5 to 3.5 from VAN's product data. In soil above pH 8 with free lime, phosphate that has dissolved meets calcium within a few centimetres and drops out of solution before the root arrives. The acidic zone around the granule slows that reaction for the weeks a young root needs to reach it. It is a zone of a few centimetres, for a few weeks. It does not change the field and it will not move a soil test.

Say it like this: "Each granule keeps its own small patch acidic for a few weeks. That is the root's window. Not the field, not the season."



The granule's own solution starts acidic. The zone is a few centimetres, not the field.

Textbook chemistry of a superphosphate granule, stated as general soil chemistry, not a VAN measurement. No percentage of phosphate fixed or saved is printed.

2 basal bags. A choice, not a contest.

At land preparation the grower may put in Fusion Phosphate or Green Phosphate; the programme's bag count is the same for either, on purpose. Fusion Phosphate is the straight bag: 23 kg of P_2O_5 , no nitrogen, no coating, 43% water soluble. Green Phosphate is the coated bag: 16 kg of P_2O_5 with 3 kg of nitrogen, granulated in an acidic mix and sealed with humic. The kilograms on the 2 bags differ. VAN prints no trial saying one reaches the crop better than the other, because none exists, and the published Pakistani field trials found no phosphate source beat another consistently.

Say it like this: "Both go in at land preparation, same number of bags. One is straight phosphate with no nitrogen. One is coated. You choose; I will tell you what is in each."



What is in each bag, from the declared analyses. No availability percentage and no “better” is printed.

What is verified for Fusion Phosphate, and what is not.

No trial of Fusion Phosphate is on VAN's record. What a dealer may state as fact is the bag, the standard, the licence, the soil record and the published soil chemistry, each as what it is.

Item	What it says	How a dealer may use it
The bag	Fusion Phosphate, triple super phosphate, P_2O_5 46%, 50 kg, granular 1 to 4 mm. 43% water-soluble P_2O_5 as van.com.pk prints it.	As fact.
The standard and licence	Pakistan Standard PS 216-2009 for triple super phosphate; manufacturing licence CM/L-2942/2023.	As fact. The number, never an expiry date.
The soil record (Punjab, 770,160 samples)	69.1% of samples above pH 8; 97.9% below adequate in available phosphorus, 59.1% low; mean 6.62 ppm. 35 of 36 districts read Critical or Weak.	As fact about the soil. Never as product performance.
The soil chemistry (published)	Above pH 8 with free lime, dissolved phosphate meets calcium within centimetres and drops out of solution; the first calcium phosphates are still fairly available, the loss is their ageing into rock-like forms.	As general soil chemistry. Not as a percentage fixed.
Pakistani calcareous-soil trials (published)	Ahmad et al. 2022; Mussarat et al. 2021; Mehdi et al. 2003: no phosphate source beat another consistently; the consistent gain came from organic matter applied with the phosphate.	As the reason "TSP beats DAP" and "TSP beats Green Phosphate" are never said.
Not on file	A Fusion Phosphate trial; a yield figure; a percentage of phosphate fixed or saved; a Pakistani figure for how far phosphate moves from a granule.	Not said.

Sources: the bag and van.com.pk; the PSQCA register as van.com.pk prints it; the Punjab soil testing programme (36 district workbooks, 770,160 samples); Tunesi 1999 and Freeman and Rowell 1981 for the calcium chemistry; Ahmad et al. 2022, Mussarat et al. 2021 and Mehdi et al. 2003 for the Pakistani trials.

The bag, in the grower’s own back-of-envelope.

Price from the VAN farmer price sheet, Aug 2026, delivered farmer price. P₂O₅ from the declared analysis. Nothing else is in the bag.

Fusion Phosphate, P ₂ O ₅ 46%	Weight	P ₂ O ₅	of it water soluble	Rupees
1 bag	50 kg	23.0 kg	21.5 kg	15,000
½ bag	25 kg	11.5 kg	10.75 kg	7,500
1.5 bags (the sunflower programme)	75 kg	34.5 kg	32.25 kg	22,500

A bag is 23 kg of P₂O₅ and nothing else: no nitrogen, so the grower sets his nitrogen with his nitrogen bag. That is the whole of the counter arithmetic. What the root meets of the 23 kg is the mechanism slide, not a number.

Fusion Phosphate 50 kg PKR 15,000: the VAN farmer price sheet, Aug 2026. 23.0 kg = 50 × 0.46; 21.5 kg = 50 × 0.43 (the water-soluble figure van.com.pk prints). Sunflower: the crop programmes on van.com.pk, 1.5 bags at land preparation.

Per acre, where the programme lets the grower choose it.

Sunflower names Fusion Phosphate outright. On every programme that names Green Phosphate, Fusion Phosphate is the grower's choice at the same bag count. Rupees at the VAN farmer price sheet, Aug 2026.

Crop	Bags per acre	P ₂ O ₅ put in	Rupees per acre	How the programme prints it
Sunflower	1.5 bags of 50 kg	34.5 kg	22,500	Fusion Phosphate, land preparation
Wheat	1 bag of 50 kg	23 kg	15,000	Green Phosphate, or Fusion Phosphate by choice
Maize	1.5 bags of 50 kg	34.5 kg	22,500	Green Phosphate, or Fusion Phosphate by choice
Cotton	1.5 bags of 50 kg	34.5 kg	22,500	Green Phosphate, or Fusion Phosphate by choice
Potato	2 bags of 50 kg	46 kg	30,000	Green Phosphate, or Fusion Phosphate by choice
Sugarcane	2 bags of 50 kg	46 kg	30,000	Green Phosphate, or Fusion Phosphate by choice
Basmati rice	1 bag of 50 kg	23 kg	15,000	Green Phosphate, or Fusion Phosphate by choice

The bag count does not change when the grower switches bag, on purpose. He changes the bag, not the plan.

Sunflower row and the Green Phosphate rows: the crop programmes on van.com.pk, where Fusion Phosphate is shown as the choice beside Green Phosphate at the same bag count. Rupees = bags × 15,000.

What a tonne of harvest takes off the field, against 1 bag.

Removal per tonne from the IPNI tables the site’s balance block uses. The grower multiplies by his own yield; VAN does not guess it for him.

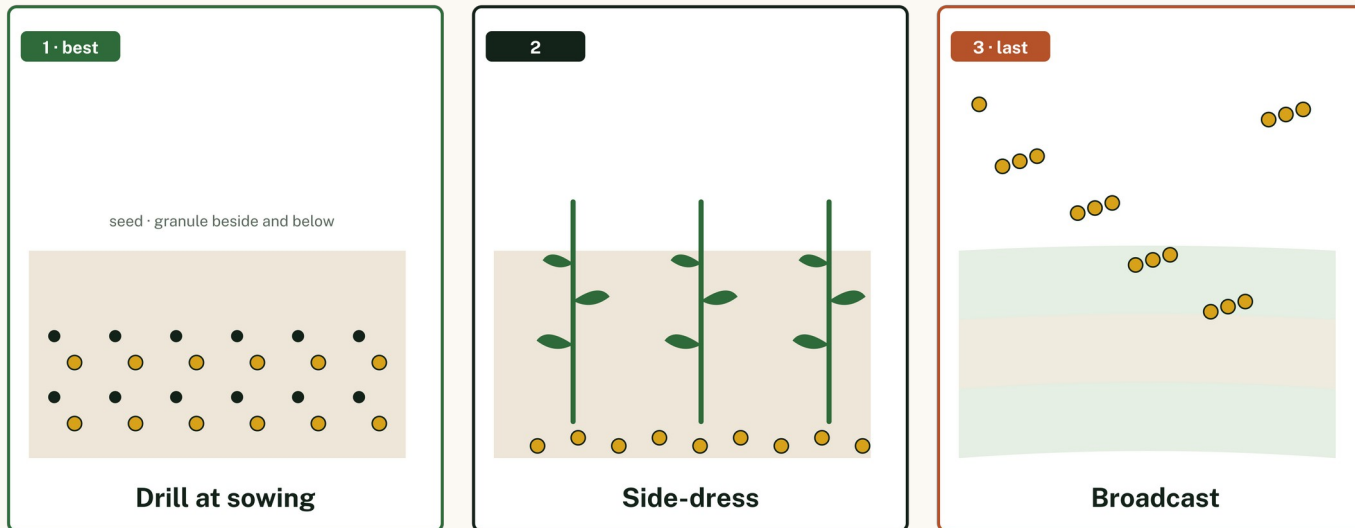
Per tonne harvested	P ₂ O ₅ in the grain or seed	P ₂ O ₅ in the straw or stover	Both together
Wheat, per tonne of grain	8.0 kg	2.7 kg	10.7 kg
Maize, per tonne of grain	6.3 kg	2.9 kg	9.2 kg
Sunflower, per tonne of seed	9.7 kg	2.4 kg	12.1 kg
1 bag of Fusion Phosphate puts in			23.0 kg

At 40 maunds of wheat (1.6 tonnes) the grain takes off 12.8 kg of P₂O₅ and the bhusa 4.3 kg more: 17.1 kg against the 23 kg in 1 bag. The account shows why the basal bag is the season’s phosphate, and why the soil, not the bag, decides how much of it the root meets.

IPNI nutrient removal tables (metric, 01/15): wheat grain P₂O₅ 8.0, straw 2.7 per tonne of grain; maize 6.3 and 2.9; sunflower 9.7 and 2.4. 40 maunds × 40 kg = 1.6 t. No Pakistani removal table exists on the public record, as van.com.pk says.

HOW IT GOES ON

Basal. Drill at sowing, beside and below the seed. Never with the seed, never in the tank.



1. Drill at sowing, best

The seed drill's fertilizer box places the granule about 5 cm beside and below the seed row, never touching the seed. The root meets it first and the whole bag is in the root zone. Phosphorus does not travel, so this is the method that matters most for this bag.

2. Side-dress, second

A band beside the row, by hand or planter, at sowing or the first irrigation, covered with soil. The sunflower programme prints side dressing as its second method.

3. Broadcast, last

Spread before the last ploughing or before an irrigation and mixed in. Most of the bag is away from the root. It is a basal granule for the soil: never a foliar, never in the fertigation tank.

Drill first, side-dress second, broadcast last.

The order for every basal granule VAN sells. The same bag gives a different result by where it lands.

Order	Method	What it means	Why
1 · best	Drill at sowing	The seed drill's fertilizer box places the granule about 5 cm beside and below the seed row, never touching the seed.	The root meets it first. Least contact with air, sun and free calcium at the surface. The whole bag is in the root zone.
2	Side-dress	A band beside the row, by hand or planter, at sowing or the first irrigation, covered with soil.	Still a band near the root, but later and shallower than a drill.
3 · last	Broadcast	Spread on the surface before the last ploughing or before an irrigation, and mixed in.	Most of the bag is away from the root, and most of it is exposed at the surface.

In VAN's own fields and partner fields, on wheat, maize and sugarcane, basal fertilizer drilled at sowing has given 7 to 9% better response than the same bag broadcast: the mean of replicated observations across regions, not a published trial.

“15,000 for a bag of TSP? DAP at my counter is cheaper and has nitrogen in it too.”

The dealer says, out loud

Write both prices on your paper; VAN prints no DAP price. Then write what is in each bag. This bag is 23 kg of P_2O_5 and nothing else. The nitrogen in a DAP bag is 9 kg, and you are already setting your nitrogen with your nitrogen bag. So you are paying for phosphate, at the highest analysis a straight phosphate carries, in a granule that starts acidic in a soil that is not. Which of the 2 the root meets more of is not a number VAN prints.

The fact behind it

- Fusion Phosphate 50 kg PKR 15,000 (VAN farmer price sheet, Aug 2026). DAP: the counter price, not on the VAN sheet.
- 23 kg P_2O_5 and 0 kg N in this bag (the declared analysis, 46%).
- DAP 18-46-0 carries 23 kg P_2O_5 and 9 kg N per 50 kg (the declared analysis of DAP).
- Pakistani field trials found no phosphate source beat another consistently (Ahmad et al. 2022; Mussarat et al. 2021; Mehdi et al. 2003).

Never say: “TSP beats DAP” or any percentage of phosphate saved. No trial supports it and VAN prints none.

“Green Phosphate or Fusion Phosphate? Just tell me which is better.”

The dealer says, out loud

Both go in at land preparation and the programme gives the same number of bags for either. Fusion Phosphate is the straight bag: 23 kg of phosphate, no nitrogen, 43% water soluble, an acidic granule. Green Phosphate is the coated bag: 16 kg of phosphate with 3 kg of nitrogen, made in an acidic mix and sealed with humic. VAN has no trial that says one beats the other, and VAN says so. You choose; I will tell you what is in each.

The fact behind it

- Fusion Phosphate is shown as the grower's choice beside Green Phosphate, same bag count (the crop programmes on van.com.pk).
- Declared analyses: P_2O_5 46 against 6-32-0; 23 kg against 16 kg + 3 kg N per 50 kg bag.
- No Fusion Phosphate or Green Phosphate field trial exists, and the Green Phosphate page says so.

Never say: “The coated one is 44% more available” or “the straight one has 44% more phosphate so it is better”. Kilograms are not the comparison.

“Can I put it through the drip, or spray it?”

The dealer says, out loud

No. It is a basal granule for the soil. Drill it at sowing beside and below the seed row, side-dress it in a band, or broadcast and mix it in before the crop. Only 43 of the 46 units dissolve in water and it is not made for a tank or a nozzle. For a soluble mid-season phosphate the programme names a different product, at its own stage.

The fact behind it

- The product page lists soil application only; the sunflower row reads drill at sowing (or broadcast), with side dressing as the alternative (van.com.pk).
- The 3 methods for a basal granule: drill, side-dress, broadcast, in that order.
- The phosphate routes on van.com.pk: Fusion Phosphate is the soil basal route; V. Ammonium Phosphate is the fertigation route.

Never say: A foliar or fertigation rate for Fusion Phosphate.

“I forgot it at sowing. Can I top-dress it now?”

The dealer says, out loud

Phosphorus does not travel to the root; the root has to grow to it. A granule on the surface of a standing crop sits where the roots are not, and on this soil the calcium reaches it before a root does. For the crop standing now, follow the programme's next stage. Plan this bag for the next sowing, drilled at land preparation.

The fact behind it

- Phosphorus damage in the first weeks cannot be undone later (van.com.pk, the 17 nutrients page).
- Broadcast is the last method because most of the bag is away from the root.
- Every phosphate route on van.com.pk puts the basal bag at sowing or land preparation.

Never say: “Top-dress it and it will catch up.”

“Will it burn the seed if I drill it?”

The dealer says, out loud

Drill it the way every basal granule from VAN is drilled: about 5 cm beside and below the seed row, never touching the seed. This bag carries no nitrogen, so there is no ammonia beside the seed from it. It is still placed beside and below, never with the seed.

The fact behind it

- Placement for every VAN basal granule: beside and below the seed, never touching the seed (the application slide).
- Declared analysis: P_2O_5 46%, no nitrogen.
- Triple super phosphate carries no ammonium, so it releases no ammonia (general chemistry).

Never say: “It is seed safe, put it in the seed row.” VAN prints no seed-safety claim.

“How many bags an acre?”

The dealer says, out loud

What the programme says for your crop. Sunflower names it: 1.5 bags at land preparation. Where the programme names Green Phosphate and you choose Fusion Phosphate instead, the bag count stays the same: 1 on wheat and rice, 1.5 on maize and cotton, 2 on potato and sugarcane. More bags than the programme is not more crop; the soil decides what the root meets.

The fact behind it

- Sunflower: 1.5 bags at land preparation (the crop programmes on van.com.pk).
- The count is unchanged on a bag switch, on purpose (van.com.pk shopping list).
- Green Phosphate rows on van.com.pk: wheat 1, maize 1.5, cotton 1.5, potato 2, sugarcane 2, basmati rice 1.

Never say: “Half a bag will do because it is 46%.”

“Is it registered? Whose licence?”

The dealer says, out loud

It is registered against Pakistan Standard PS 216-2009, the standard for triple super phosphate, under manufacturing licence CM/L-2942/2023, VAN as the maker. Every batch is tested at VAN Lab, PNAC LAB 336, before it ships. Read the batch number off your bag and VAN sends you that batch's certificate, free.

The fact behind it

- Fusion Phosphate: PS 216-2009 (TSP), CM/L-2942/2023 (the regulatory table on van.com.pk).
- Every batch is released through VAN Lab, PNAC LAB 336 (the corporate profile).
- The certificate of analysis for any batch is free on request, by batch number.

Never say: An expiry date for the licence, or “government approved” as a claim about performance.

“My soil test says phosphorus is low and I put a bag on every year. Why?”

The dealer says, out loud

Because the field is calcareous. Above pH 8 with free lime, phosphate that dissolves meets calcium within a few centimetres and drops out of solution, and over the years it ages into forms the root cannot use. A soil test measures the field. The acidic zone around each Fusion Phosphate granule is a few centimetres for a few weeks: it gives the root its window, and it does not change the reading on your report. 97.9% of Punjab samples read below adequate in phosphorus, so your report has company.

The fact behind it

- 69.1% of Punjab samples read above pH 8; 97.9% below adequate in available phosphorus (the Punjab soil record, 770,160 samples).
- Above pH 8 with free lime, dissolved phosphate meets calcium within centimetres and drops out of solution (Tunesi 1999; Freeman and Rowell 1981).
- The acidic zone is a microsite, not a soil-test change (the Green Phosphate page on van.com.pk).

Never say: “It raises your soil phosphorus” or “80% of your phosphate is lost”.

Fusion Phosphate in the published programmes, per acre.

Crop	Stage	Rate per acre	Method	Pack
Sunflower	Land Preparation	1.5 bag	Drill at sowing; broadcast if no drill	50 kg

WHAT VAN WILL NOT CLAIM FOR IT

A dealer who says any of these is speaking for himself, not for VAN.

1 A yield figure, or that TSP beats DAP or Green Phosphate. No Fusion Phosphate trial exists, and the Pakistani field trials found no source beat another consistently.

2 That 80% of applied phosphate is fixed, or any percentage fixed or saved. VAN does not print that line.

3 A soil-test change or a field-scale pH change. The acidic zone is a few centimetres around each granule for a few weeks.

4 A foliar or fertigation use or rate. It is a basal granule for the soil.

5 That fewer, heavier bags mean more crop. Concentration is a freight fact, not a crop result.

6 A licence expiry date, or “government approved” as a performance claim. The dealer quotes the number.

Every batch is tested. The bag carries the proof.

1 · The batch number

It is printed on the bag. Read it to the grower and show him where it is. Every bag of every batch carries one.

2 · The certificate of analysis

For any batch, free on request. Send the batch number off the bag to kisan@van.com.pk or the helpline, and the certificate for that batch comes back. No sample to post.

3 · The laboratory

VAN Lab, PNAC LAB 336, accredited to ISO/IEC 17025:2017. Every batch is released through it before it ships. The certificate carries the laboratory's number.

What a certificate for Fusion Phosphate shows

The declared analysis tested for the batch: total P_2O_5 against 46%, with the water-soluble fraction where the standard requires it, the batch number, the Pakistan Standard PS 216-2009 and licence CM/L-2942/2023 on the bag. No certificate for a Fusion Phosphate batch was on file for this deck, so no batch reading is printed here; the grower asks for his own batch's certificate.

The lines a dealer says in Urdu.

The first line is the promise word from the VAN brand master. The other 3 are the deck's key sentences, translated in a first pass and not yet reviewed by VAN. Say them, and send corrections to partner@van.com.pk.

REACH. The promise word for phosphorus.

Urdu first pass, unreviewed

پہلے جڑ، پھر سب کچھ

Phosphorus does not travel to the root. The root has to reach it.

Urdu first pass, unreviewed

فاسفورس جڑ تک نہیں چلتا، جڑ کو اس تک پہنچنا پڑتا ہے

Drill it at sowing, beside and below the seed, never with the seed.

Urdu first pass, unreviewed

بوائی پر ڈرل کریں، بیج کے پاس اور نیچے، بیج کے ساتھ کبھی نہیں

Do not compare kilograms. Compare what reaches the crop.

Urdu first pass, unreviewed

کلو کا موازنہ نہ کریں، وہ دیکھیں جو فصل تک پہنچے

Write to the address for what you need.

If you grow

kisan@van.com.pk

A crop plan, product rates, a certificate for your batch, a dealer near you.

If you sell

partner@van.com.pk

Dealers, distributors and the trade. Stock, training, corrections to this deck.

Everything else

info@van.com.pk

Supply, the laboratory, a plant visit.

Helpline 0300 5003041 · www.van.com.pk

Vital Agri Nutrients (Pvt) Ltd, Lahore. 1 helpline, 3 addresses. No group broadcasts: a grower who writes gets 1 reply.

Fusion Phosphate: in the ground at sowing, drilled beside and below the seed. Do not compare kilograms; compare what reaches the crop.

